

Experiment 5:

Write a C Program to Implement Bully Election Algorithm.

Aim

To implement the Bully Election Algorithm for selecting a coordinator in a distributed system.

Objective

- To understand leader election in distributed systems
- To simulate failure detection and recovery
- To implement process coordination using the Bully Algorithm

Theory

The Bully Election Algorithm is used in distributed systems to elect a coordinator (leader) when the current coordinator fails.

Key Concepts:

- Each process has a unique ID
- The process with the highest ID becomes the coordinator
- When a process detects failure:
 1. It sends election messages to all higher-ID processes
 2. If no response → it becomes coordinator
 3. If response received → higher process continues election

Algorithm

1. Start
2. Input number of processes
3. Assign status (Active/Failed) to each process
4. Select process that detects failure
5. Send election message to higher processes
6. If higher process responds → it takes over
7. Repeat until highest active process is found
8. Declare coordinator
9. Stop

Program (C Code)

```
#include <stdio.h>

int processes[10], n;
int coordinator;

void election(int initiator)
{
    int i;
    printf("\nProcess %d is initiating election...\n", initiator);

    for (i = initiator + 1; i < n; i++)
    {
        if (processes[i] == 1)
        {
            printf("Process %d sends election message to Process %d\n", initiator, i);
            election(i);
        }
    }
    coordinator = initiator;
}

void displayCoordinator()
{
    printf("\nCurrent Coordinator is Process %d\n", coordinator);
}

int main()
{
    int i, failed;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter status of each process (1 = Active, 0 = Failed):\n");
    for (i = 0; i < n; i++)
    {
        printf("Process %d: ", i);
        scanf("%d", &processes[i]);
    }
    printf("\nEnter the process which detects failure: ");
    scanf("%d", &failed);
    election(failed);
    displayCoordinator();

    return 0;
}
```

Result

The program successfully simulates the Bully Election Algorithm and elects the highest active process as the coordinator.

Conclusion

The Bully Algorithm is an efficient method for coordinator selection in small distributed systems, though it may not scale well due to message overhead.

FAQ:

1. **What are the advantages/ disadvantages of Bully Algorithm?**
2. **What happens when multiple processes start election simultaneously?**
3. **What happens when a failed process recovers?**
4. **Can more than one coordinator exist at the same time?**
5. **How does the algorithm ensure no starvation?**
6. **What type of system is required for Bully Algorithm to work?**